Numerical Experiments: Large case

3 Large Instance: City-Specific Market Size and Random Choice Probabilities

Purpose. Apply the model to an annual instance whose inputs follow the parameter rationale document (`4. References/parameter_rationale_baseline.tex`): three candidate cities, a fleet of 1,000 vehicles, and ten demand scenarios. The instance is synthetic: the inputs are literature-based proxies, not calibrated data for real cities. The questions are the same as for the small synthetic instance (Section 2): does the implementation match the formulation, which inputs change the optimal allocation, and where are the switching points.

Two design choices differ from the small instance. First, market size M_c differs across cities (0.75, 1, and 1.25 times the reference value), while all cost and productivity inputs are identical across cities. Second, the choice probabilities p_c^s are not designed by hand but drawn at random, independently for every city and scenario, from four levels. The levels $\{0.002, 0.005, 0.01, 0.02\}$ are chosen so that demand and capacity are of the same order of magnitude. With the reference levels $\{0.02, 0.1, 0.2, 0.4\}$ the smallest scenario demand of any city (11.6 million trips) already exceeded a city's maximum capacity (4.38 million trips); every vehicle then serves κ trips wherever it is placed, and any two cities with 500 vehicles each are optimal regardless of M_c and p_c^s. Because $D_c^s = M_c p_c^s$, using levels one tenth as large is identical to scaling every M_c by 0.1.

3.1 Baseline Experiment

3.1.1 Inputs and Common Settings

Table 1: Baseline Inputs by City

Parameter	Definition and unit	A	B	C	Basis (see rationale document)
M_c	Potential trip opportunities (trips/year)	577,908,105	770,544,140	963,180,175	0.75, 1, 1.25 × 770,544,140 (= 38,527,207/0.05); proxy value
κ_c	Service capacity (trips/vehicle/year)	8,760	8,760	8,760	24 × 365
$\bar{q}_c$	City-specific vehicle limit (vehicles)	500	500	500	Assumption
F_c	Initial entry cost (USD/city)	3,500,000	3,500,000	3,500,000	(300,000 + 2,500,000) × 1.25
K_c	Fixed vehicle cost (USD/vehicle/year)	7,972	7,972	7,972	4,334 + 1,131 + 1,694 + 813
c_c^+	Unused-capacity penalty (USD/trip)	0	0	0	No additional idle loss (assumption)
c_c^-	Forgone contribution margin (USD/trip)	10	10	10	Revenue 18 minus variable cost 8

Table 2: Common Settings

Item	Setting
C, Q	$C = \{A, B, C\}$, total fleet limit $Q = 1{,}000$ vehicles
S	$\lvert S\rvert = 10$, probability of each scenario $1/\lvert S\rvert = 0.1$
p_c^s	Drawn independently for each (c, s) from $\{0.002, 0.005, 0.01, 0.02\}$ with equal probability (random seed 2026)
D_c^s	$D_c^s = M_c p_c^s$
Evaluation period	One year after entry: one-time entry cost plus costs and losses for that year
y_c, q_c	Chosen before observing demand and applied to every scenario
Operations	No vehicle relocation or capacity sharing across cities; demand in cities not entered is counted as unmet
Verification	SciPy MILP (HiGHS) compared with an independent dynamic program over integer vehicle counts, which counts all tied optima; the dynamic program was first checked against full enumeration on the small instance (eight cases, baseline objective 858 reproduced)

Each row of Table 3 is one joint demand scenario. Because the levels are drawn independently, a scenario can combine high demand in one city with low demand in another.

Table 3: Random Choice Probabilities and Demand Scenarios (Trips/Year)

s	p_A^s	p_B^s	p_C^s	D_A^s	D_B^s	D_C^s
1	0.02	0.002	0.002	11,558,160	1,541,088	1,926,360
2	0.01	0.005	0.005	5,779,081	3,852,721	4,815,901
3	0.002	0.005	0.01	1,155,816	3,852,721	9,631,802
4	0.005	0.02	0.02	2,889,541	15,410,883	19,263,600
5	0.01	0.02	0.01	5,779,081	15,410,883	9,631,802
6	0.002	0.02	0.01	1,155,816	15,410,883	9,631,802
7	0.002	0.005	0.002	1,155,816	3,852,721	1,926,360
8	0.02	0.01	0.02	11,558,160	7,705,441	19,263,600
9	0.005	0.01	0.01	2,889,541	7,705,441	9,631,802
10	0.02	0.002	0.01	11,558,160	1,541,088	9,631,802
Mean	0.0096	0.0099	0.0099	5,547,917	7,628,387	9,535,483

3.1.2 Demand Scale versus Capacity

Before optimizing, compare demand with the capacity a city can deploy at most, $\kappa_c \bar{q}_c = 4{,}380{,}000$ trips. The last column gives the choice probability at which demand equals this capacity.

Table 4: Scenario Demand and Maximum City Capacity (Trips/Year)

c	$\min_s D_c^s$	$\bar{D}_c$	$\max_s D_c^s$	$\kappa_c \bar{q}_c$	Scenarios with $D_c^s > \kappa_c \bar{q}_c$	$\kappa_c \bar{q}_c / M_c$
A	1,155,816	5,547,918	11,558,162	4,380,000	5/10	0.0076
B	1,541,088	7,628,387	15,410,883	4,380,000	5/10	0.0057
C	1,926,360	9,535,484	19,263,604	4,380,000	8/10	0.0045

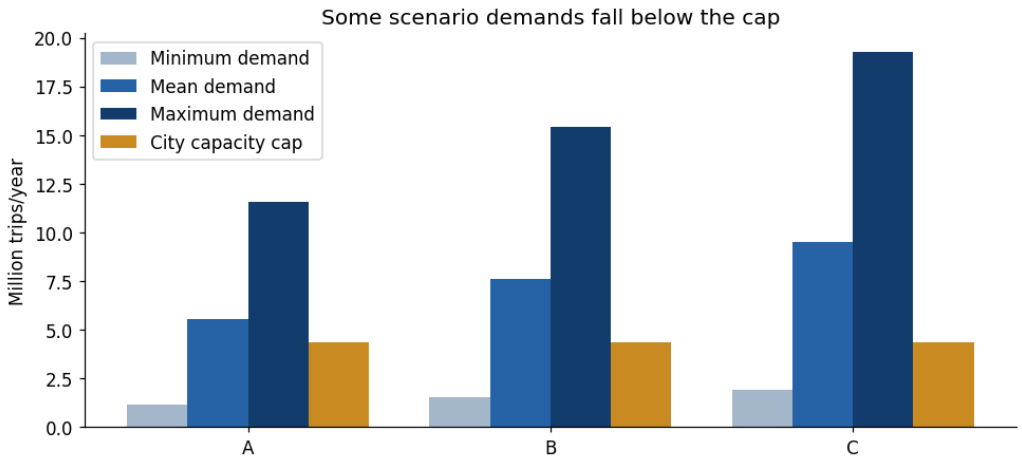


Figure 1: Minimum, mean, and maximum scenario demand compared with the maximum city capacity.

Interpretation. Demand straddles capacity: in half of the scenarios for A and B, and in eight of ten for C, demand exceeds what 500 vehicles can serve, while in the remaining scenarios part of the capacity would be idle. This is the regime in which the allocation depends on demand. Meeting mean demand in every city would require 2,593 vehicles against a fleet of 1,000.

3.1.3 Optimal Allocation and Cost

The optimal solution enters all three cities. The representative allocation is $q = (132, 368, 500)$, but it is not unique: 72 allocations share the same objective. In all of them $q_A = 132$, while q_B ranges from 368 to 439 and q_C from 429 to 500 with $q_B + q_C = 868$. All checks passed: integrality, nonnegativity, city-specific and total fleet limits, demand–capacity balance, objective recalculation, and agreement between the MILP and the dynamic program.

Table 5: City-Level Results for the Representative Optimum (Trips/Year)

c	y_c	q_c	$\kappa_c q_c$	$\bar{D}_c$	$\bar{T}_c$	$\bar{U}_c$	$\bar{O}_c$
A	1	132	1,156,320	5,547,918	1,156,169	4,391,749	151
B	1	368	3,223,680	7,628,387	2,887,162	4,741,225	336,518
C	1	500	4,380,000	9,535,484	3,889,272	5,646,212	490,728
Total	3	1000	8,760,000	22,711,789	7,932,603	14,779,186	827,397

Table 6: Baseline Cost Breakdown and Performance (USD)

C_F	C_K	C_O	C_U	Z	Z_0	ΔZ
10,500,000	7,972,000	0	147,791,859	166,263,859	227,117,885	60,854,026

The service rate is SR = 34.93% and capacity utilization is CU = 90.55%. On average 827,397 trips of capacity are unused, almost all of it in B and C: in the scenarios where their demand is at the lowest level, the deployed capacity exceeds demand. Subtracting entry costs of 10.5 million and vehicle costs of 7.97 million from the service contribution of $7{,}932{,}603 \times 10 = 79.3$ million gives an improvement of 60.85 million.

Table 7: Scenario Totals for Allocation $(132, 368, 500)$ (Trips/Year)

s	$\sum_c D_c^s$	$\sum_c T_c^s$	$\sum_c U_c^s$	$\sum_c O_c^s$
1	15,025,610	4,623,769	10,401,840	4,136,231
2	14,447,700	8,760,000	5,687,703	0
3	14,640,340	8,759,496	5,880,842	504
4	37,564,030	8,760,000	28,804,030	0
5	30,821,770	8,760,000	22,061,770	0
6	26,198,500	8,759,496	17,439,000	504
7	6,934,897	6,305,857	629,041	2,454,143
8	38,527,210	8,760,000	29,767,210	0
9	20,226,780	8,760,000	11,466,780	0
10	22,731,050	7,077,408	15,653,640	1,682,592
Mean	22,711,789	7,932,603	14,779,186	827,397

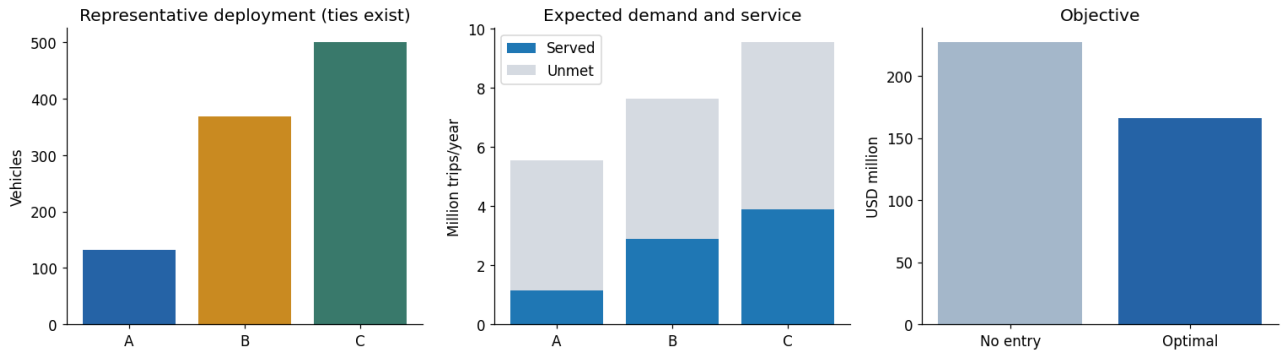


Figure 2: Representative allocation, expected served and unmet demand by city, and the objective compared with no entry.

3.1.4 Why 132 Vehicles in A, and Why B and C Tie

The allocation can be read from the value of one additional vehicle. A vehicle adds $\kappa = 8{,}760$ trips of capacity and costs $K = 7{,}972$. In a scenario where demand still exceeds capacity, it serves 8,760 more trips and avoids $8{,}760 \times 10 = 87{,}600$ of forgone margin; in a scenario where demand is already fully served it adds nothing, because $c^+ = 0$. The expected value of one more vehicle is therefore

$$87{,}600 \times \big(\text{share of scenarios with unmet demand}\big) - 7{,}972.$$

As vehicles are added to a city, the low-demand scenarios become fully served one after another and this value steps down (Table 8).

Table 8: Value of the k-th Vehicle in Each City (USD/Year)

c	Vehicle k	Scenarios with unmet demand	Value of vehicle k
A	1–131	10/10	79,628
A	132	boundary	78,117
A	133–329	7/10	53,348
A	330	boundary	50,829
A	331–500	5/10	35,828
B	1–175	10/10	79,628
B	176	boundary	78,285
B	177–439	8/10	62,108
B	440	boundary	57,070
B	441–500	5/10	35,828
C	1–219	10/10	79,628
C	220	boundary	77,949
C	221–500	8/10	62,108

The optimizer fills the most valuable slots first: A's first 132, B's first 176, and C's first 220 vehicles, 528 in total, each worth 77,949 or more. The remaining 472 vehicles go to the slots worth 62,108, B's 177th–439th and C's 221st–500th; these 543 slots have identical value, so any split of the 472 vehicles between B and C is optimal. This produces the 72 tied allocations. A's next vehicle is worth 53,348, less than 62,108, so A stays at 132. The ties are a consequence of ten equally likely scenarios: the value of a vehicle moves in steps of one tenth, and B and C sit on the same step over a wide range of vehicle counts.

Entering A is worthwhile, but only narrowly. Table 9 compares the optimum with other allocations of 1,000 vehicles. A's 132 vehicles contribute 10.51 million before entry cost; the entry cost is 3.5 million; and the same 132 vehicles placed in B (its 369th–500th, of which 60 serve only five scenarios) would contribute 6.62 million. The net advantage of entering A is $10.51 - 3.50 - 6.62 = 0.39$ million. This margin determines the switching points in Section 3.3.

Table 9: Comparison of Allocations Using 1,000 Vehicles (Costs in USD Million)

q	$\bar{T}$ (trips)	C_F	C_K	C_U	Z	ΔZ
$(132, 368, 500)$	7,932,603	10.50	7.97	147.79	166.26	60.85
$(132, 439, 429)$	7,932,603	10.50	7.97	147.79	166.26	60.85
$(0, 500, 500)$	7,543,306	7.00	7.97	151.68	166.66	60.46
$(334, 333, 333)$	7,748,391	10.50	7.97	149.63	168.11	59.01
$(500, 250, 250)$	7,312,143	10.50	7.97	154.00	172.47	54.65
$(500, 0, 500)$	7,003,925	7.00	7.97	157.08	172.05	55.07
$(500, 500, 0)$	6,768,687	7.00	7.97	159.43	174.40	52.71

Interpretation. The first two rows are two of the 72 tied optima. Leaving A out costs 0.39 million; an even split across cities costs 1.84 million more than the optimum because it moves vehicles from B and C's 62,108-valued slots into A's 53,348- and 35,828-valued slots. Concentrating in two cities without C is worst: C has the largest market and the fewest low-demand scenarios.

3.2 Edge-Case Experiments

Each experiment changes only the listed input relative to the baseline. Because the expected properties of the small instance were tied to its numbers, the checks here are stated as properties that hold regardless of input scale (no entry, local optimality, or bounds on the improvement).

Table 10: Modified Inputs and Expected Properties for Edge Cases

Case	Modified input	Expected property
E1	$Q = 0$	$y = q = 0$; $U_c^s = D_c^s$
E2	$M = (0, 0, 0)$	$y = q = 0$; $Z = 0$, $U_c^s = O_c^s = 0$
E3	$\bar{q} = (0, 500, 500)$	$y_A = q_A = 0$; B, C locally optimal
E4	$c^- = (0, 0, 0)$	No service incentive: $y = q = 0$, $Z = 0$
E5	$F_c = Z_0 + 1$ for all c	Entry cost above total service value: $y = q = 0$
E6	$Q = 2{,}000$	Fleet limit not binding: $\sum_c q_c \leq 1{,}500$, $\Delta Z \geq$ baseline
E7	$F = (0, 0, 0)$	$\Delta Z \geq$ baseline; $\sum_c q_c \leq Q$
E8	$p_c^s = \kappa_c / M_c$ $(\forall c, s)$, $F = 0$	$D_c^s = 8{,}760$; $q = (1, 1, 1)$; $U_c^s = O_c^s = 0$

Table 11: Allocation and Service Results for Edge Cases (Trips/Year)

Case	q^{rep}	Ties	$\bar{T}$	SR (%)	$\bar{U}$	$\bar{O}$	Check
E1	$(0, 0, 0)$	1	0	0.00	22,711,790	0	PASS
E2	$(0, 0, 0)$	1	0	—	0	0	PASS
E3	$(0, 500, 500)$	1	7,543,306	33.21	15,168,480	1,216,694	PASS
E4	$(0, 0, 0)$	1	0	0.00	22,711,790	0	PASS
E5	$(0, 0, 0)$	1	0	0.00	22,711,790	0	PASS
E6	$(500, 500, 500)$	1	10,657,960	46.93	12,053,830	2,482,041	PASS
E7	$(132, 368, 500)$	72	7,932,603	34.93	14,779,190	827,397	PASS
E8	$(1, 1, 1)$	1	26,280	100.00	0	0	PASS

Table 12: Edge-Case Cost Breakdown and Improvement (USD Million)

Case	C_F	C_K	C_O	C_U	Z	Z_0	ΔZ
E1	0.00	0.00	0.00	227.12	227.12	227.12	0.00
E2	0.00	0.00	0.00	0.00	0.00	0.00	0.00
E3	7.00	7.97	0.00	151.68	166.66	227.12	60.46
E4	0.00	0.00	0.00	0.00	0.00	0.00	0.00
E5	0.00	0.00	0.00	227.12	227.12	227.12	0.00
E6	10.50	11.96	0.00	120.54	143.00	227.12	84.12
E7	0.00	7.97	0.00	147.79	155.76	227.12	71.35
E8	0.00	0.02	0.00	0.00	0.02	0.26	0.24

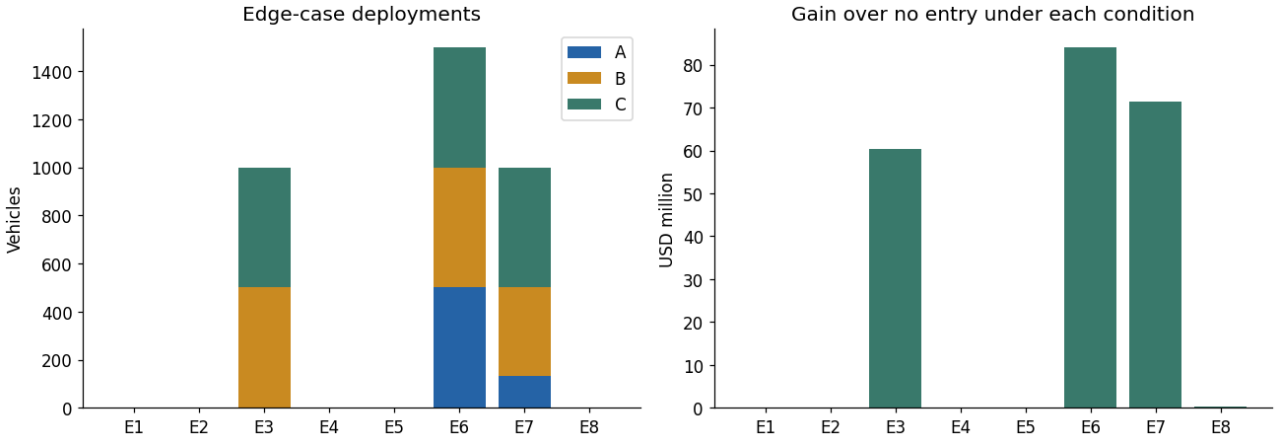


Figure 3: Allocations and improvements for E1–E8. Objectives under different conditions do not constitute a performance ranking.

Interpretation. E1, E4, and E5 result in no entry; E4 has a zero objective because unmet demand carries no loss. E3 (A cannot be entered) loses exactly the 0.39 million value of entering A. E6 relaxes the fleet limit and deploys 1,500 vehicles, the sum of the city limits, raising the improvement to 84.12 million. E7 removes entry costs but keeps the allocation, because the fleet limit, not the entry cost, caps A at 132 vehicles; the improvement rises by exactly $3F = 10.5$ million. E8 sets demand equal to one vehicle's capacity in every scenario and serves it exactly with one vehicle per city.

3.3 Sensitivity Analysis

3.3.1 Experimental Design and Interpretation

One input type is varied at a time, with all other inputs fixed at their baseline values, and the model is reoptimized. Common-level experiments change the input in all cities at once; the M_A experiment changes only A, and Section 3.4 changes one input in one city at a time.

Table 13: Parameter Ranges for the Sensitivity Experiments

Parameter	Variation and range	Note
Q	Total fleet limit: 0–2,000, step 100	$\bar{q}_c = 500$
F	Common across cities: ±25% of 3.5 million, 9 points	
K	Common across cities: ±25% of 7,972, 9 points	
κ	Common across cities: ±25% of 8,760, 9 points	
M multiplier	All cities scaled by 0.75–1.25, 9 points; city ratios 0.75:1:1.25 kept	Identical to scaling all p_c^s
p multiplier	All p_c^s scaled by 0.75, 1, 1.25	Identical to the M multiplier
$\bar{q}$	Common city limit: 100, 250, 500, 750, 1,000, 1,500	$Q = 1{,}000$
c^+	Common across cities: 0, 1, 3, 5, 10, 15, 40	Baseline 0
c^-	Common across cities: 2, 5, 10, 15, 20	
M_A	A only: 0.75–1.25 times baseline, 9 points	M_B, M_C fixed
α	Demand variability multiplier $p_c^s(\alpha) = \bar{p}_c + \alpha(p_c^s - \bar{p}_c)$: 0–1, step 0.1	City mean demands unchanged

Read the panels of each figure in the order allocation, service rate, cost breakdown, and improvement. Shaded bands show the range of each city's vehicle count across tied optimal solutions, not confidence intervals; they are drawn only at the grid points. The representative solution minimizes the number of entered cities first and then uses lexicographic order on q. Switching points are located only between adjacent grid values unless a closed-form comparison is available. In the tables, $\bar{T}$ is in trips/year and Z, ΔZ in USD million.

3.3.2 Total Fleet Limit Q

Table 14: Selected Results: Total Fleet Limit

Value	q^{rep}	Ties	$\bar{T}$	SR (%)	Z	ΔZ
0	$(0, 0, 0)$	1	0	0.00	227.12	0.00
100	$(0, 0, 100)$	3	876,000	3.86	222.66	4.46
300	$(0, 0, 300)$	1	2,487,672	10.95	208.13	18.99
500	$(0, 0, 500)$	1	3,889,272	17.12	195.71	31.41
600	$(0, 176, 424)$	205	4,898,290	21.57	189.92	37.20
800	$(0, 300, 500)$	140	6,299,890	27.74	177.50	49.62
900	$(0, 400, 500)$	40	7,000,690	30.82	171.29	55.83
1,000	$(132, 368, 500)$	72	7,932,603	34.93	166.26	60.85
1,100	$(160, 440, 500)$	1	8,608,371	37.90	160.30	66.81
1,200	$(260, 440, 500)$	1	9,221,571	40.60	154.97	72.15
1,300	$(330, 470, 500)$	31	9,781,959	43.07	150.16	76.96
1,400	$(400, 500, 500)$	61	10,219,959	45.00	146.58	80.54
1,500	$(500, 500, 500)$	1	10,657,959	46.93	143.00	84.12
2,000	$(500, 500, 500)$	1	10,657,959	46.93	143.00	84.12

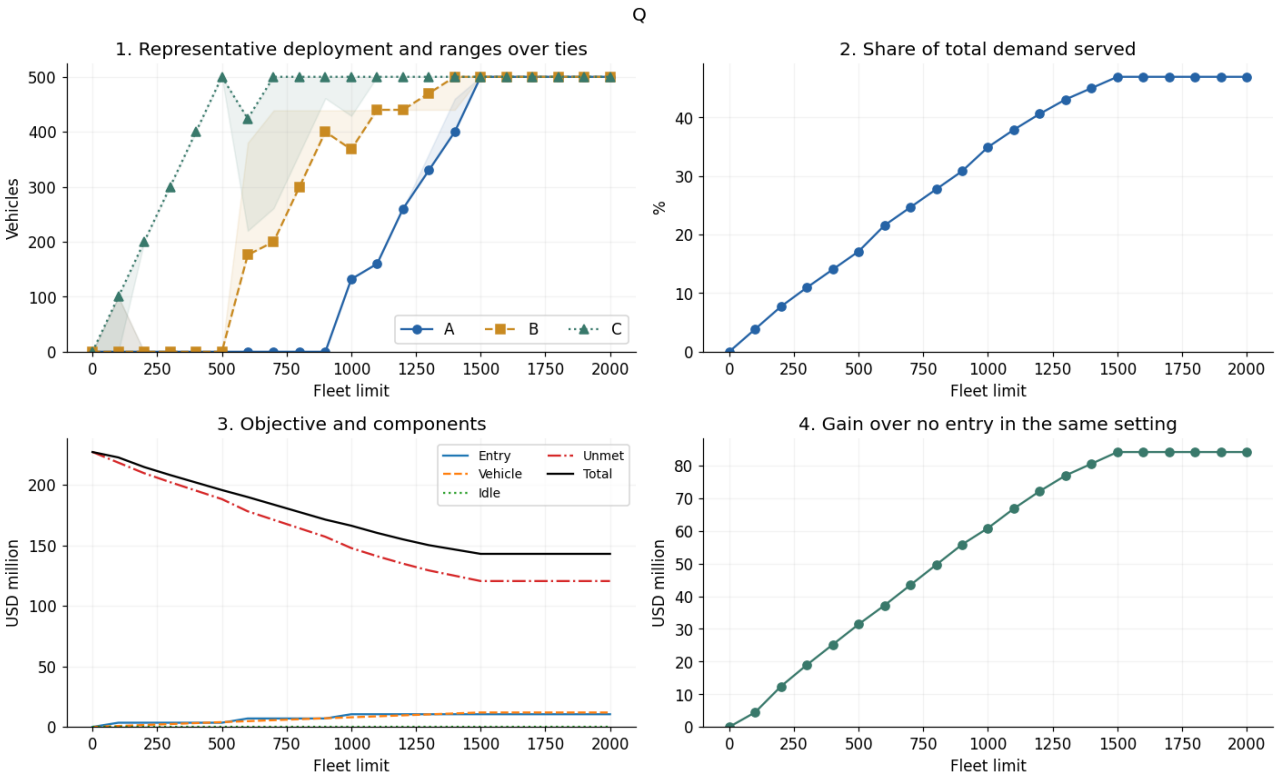


Figure 4: Allocation, service rate, cost, and improvement as the total fleet limit varies.

Interpretation. Up to 500 vehicles, everything goes to a single city, C, because a second entry cost is not yet worth paying (at $Q = 100$ any one city would do). From 600 vehicles B is entered and B and C share the fleet with many ties. A is entered only at $Q \geq 1{,}000$, and from 1,100 its fleet grows quickly because its next slots (53,348 and 35,828) are now the best remaining. At $Q \geq 1{,}500$ the city limits bind and the results stop changing.

3.3.3 Common Entry Cost F

Table 15: Selected Results: Common Entry Cost

Value	q^{rep}	Ties	$\bar{T}$	SR (%)	Z	ΔZ
2,625,000	$(132, 368, 500)$	72	7,932,603	34.93	163.64	63.48
2,843,750	$(132, 368, 500)$	72	7,932,603	34.93	164.30	62.82
3,062,500	$(132, 368, 500)$	72	7,932,603	34.93	164.95	62.17
3,281,250	$(132, 368, 500)$	72	7,932,603	34.93	165.61	61.51
3,500,000	$(132, 368, 500)$	72	7,932,603	34.93	166.26	60.85
3,718,750	$(132, 368, 500)$	72	7,932,603	34.93	166.92	60.20
3,937,500	$(0, 500, 500)$	1	7,543,306	33.21	167.53	59.59
4,156,250	$(0, 500, 500)$	1	7,543,306	33.21	167.97	59.15
4,375,000	$(0, 500, 500)$	1	7,543,306	33.21	168.41	58.71

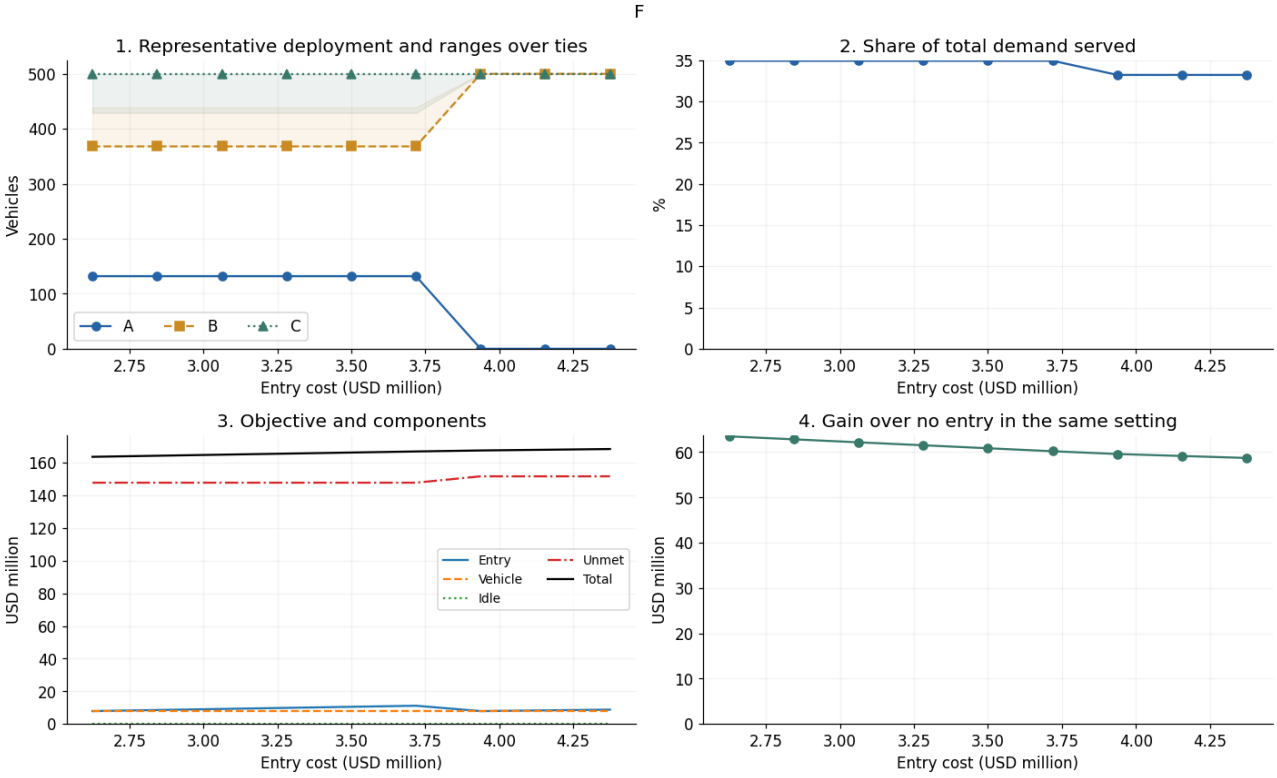


Figure 5: Results from varying the common entry cost within ±25%.

Interpretation. The entry cost changes only whether A is entered. The three-city allocation beats $(0, 500, 500)$ by 0.39 million at $F = 3.5$ million, and both alternatives pay F for B and C, so the two tie at $F = 3{,}500{,}000 + 392{,}967 = 3{,}892{,}967$; A is dropped at the next grid point, 3.94 million. Within each regime ΔZ falls linearly with F (slope -3 with three cities, -2 with two).

3.3.4 Common Vehicle Cost K

Table 16: Selected Results: Common Vehicle Cost

Value	q^{rep}	Ties	$\bar{T}$	SR (%)	Z	ΔZ
5,979.0	$(132, 368, 500)$	72	7,932,603	34.93	164.27	62.85
6,975.5	$(132, 368, 500)$	72	7,932,603	34.93	165.27	61.85
7,972.0	$(132, 368, 500)$	72	7,932,603	34.93	166.26	60.85
8,968.5	$(132, 368, 500)$	72	7,932,603	34.93	167.26	59.86
9,965.0	$(132, 368, 500)$	72	7,932,603	34.93	168.26	58.86

Interpretation. Within $\pm 25\%$ the allocation does not change: the fleet limit binds, all 1,000 vehicles are used, and a common K lowers every vehicle's value by the same amount. Only the improvement changes, by $-1{,}000$ USD per USD of K.

3.3.5 Vehicle Productivity κ

Table 17: Selected Results: Vehicle Productivity

Value	q^{rep}	Ties	$\bar{T}$	SR (%)	Z	ΔZ
6,570.0	$(0, 500, 500)$	1	5,949,490	26.20	182.59	44.52
7,117.5	$(0, 500, 500)$	1	6,387,490	28.12	178.21	48.90
7,665.0	$(0, 500, 500)$	1	6,825,490	30.05	173.83	53.28
8,212.5	$(0, 500, 500)$	1	7,187,431	31.65	170.22	56.90
8,760.0	$(132, 368, 500)$	72	7,932,603	34.93	166.26	60.85
9,307.5	$(124, 376, 500)$	38	8,370,316	36.85	161.89	65.23
9,855.0	$(118, 391, 491)$	3	8,805,480	38.77	157.54	69.58
10,402.5	$(130, 370, 500)$	38	9,188,467	40.46	153.71	73.41
10,950.0	$(148, 352, 500)$	61	9,571,761	42.14	149.87	77.25

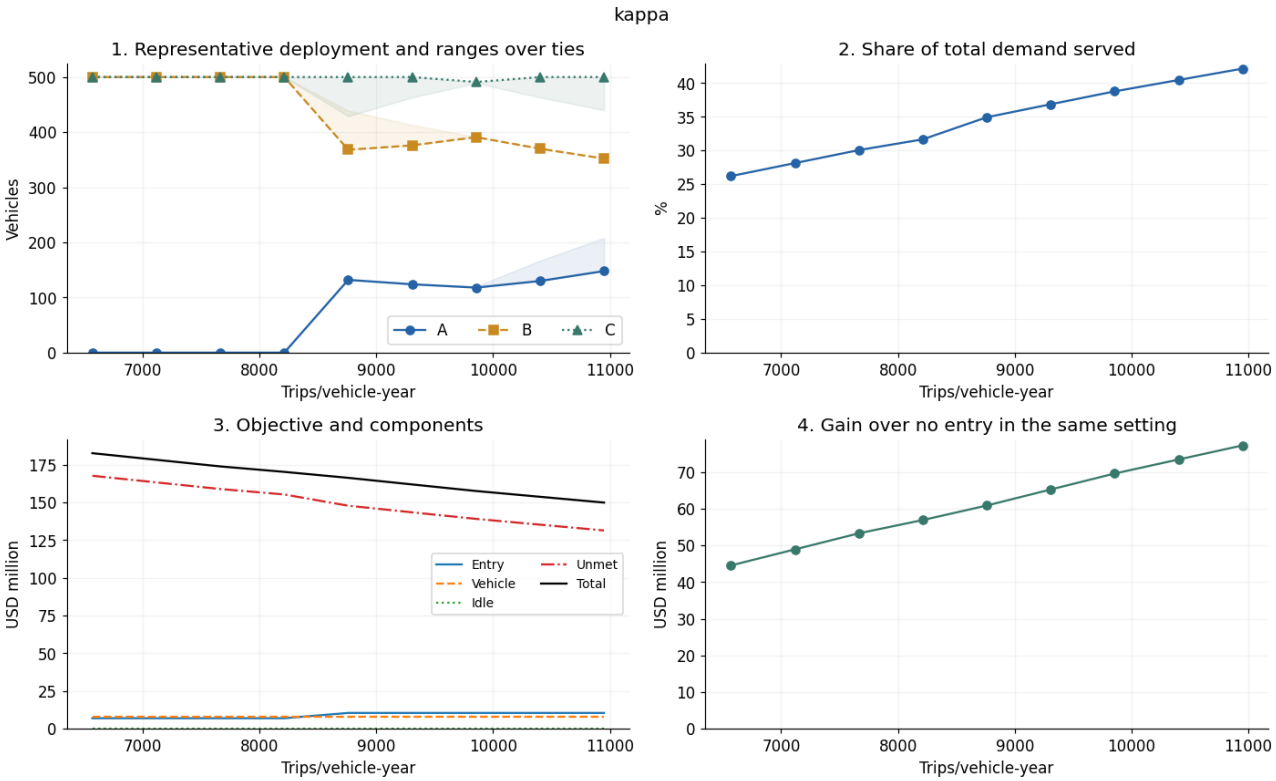


Figure 6: Results from varying service capacity per vehicle.

Interpretation. Lower productivity removes A: at $\kappa \leq 8{,}212.5$ the optimum is $(0, 500, 500)$. Two effects work against A. A's vehicles are used in all ten scenarios, so each loses 10 USD per trip of lost capacity, whereas the B vehicles they displace are used in eight or five scenarios and lose less. In addition, more vehicles are needed to cover B's 3.85-million scenarios, so B's cheap tail (vehicles serving only five scenarios) shrinks and the displaced vehicles become more valuable. The switch lies between 8,212.5 and 8,760; bisection on the same model places it near $\kappa \approx 8{,}498$. Higher productivity keeps three cities and raises the improvement by 3.8–4.4 million per grid step (6.25% of κ).

3.3.6 Market Size and Choice Probability Multipliers

Scaling all M_c and scaling all p_c^s by the same factor give identical results because only $D_c^s = M_c p_c^s$ enters the model; the p-multiplier points 0.75, 1, 1.25 coincide with the corresponding rows below.

Table 18: Selected Results: Common Multiplier on M_c (City Ratios Kept)

Value	q^{rep}	Ties	$\bar{T}$	SR (%)	Z	ΔZ
0.7500	$(170, 330, 500)$	78	7,562,071	44.39	113.19	57.15
0.8125	$(143, 357, 500)$	54	7,681,215	41.63	126.19	58.34
0.8750	$(116, 385, 499)$	18	7,800,416	39.25	139.20	59.53
0.9375	$(124, 376, 500)$	37	7,874,596	36.98	152.65	60.27
1.0000	$(132, 368, 500)$	72	7,932,603	34.93	166.26	60.85
1.0625	$(0, 500, 500)$	1	7,658,888	31.74	179.70	61.62
1.1250	$(0, 500, 500)$	1	7,774,469	30.43	192.73	62.77
1.1875	$(0, 500, 500)$	1	7,831,519	29.04	206.36	63.34
1.2500	$(0, 500, 500)$	1	7,874,862	27.74	220.12	63.78

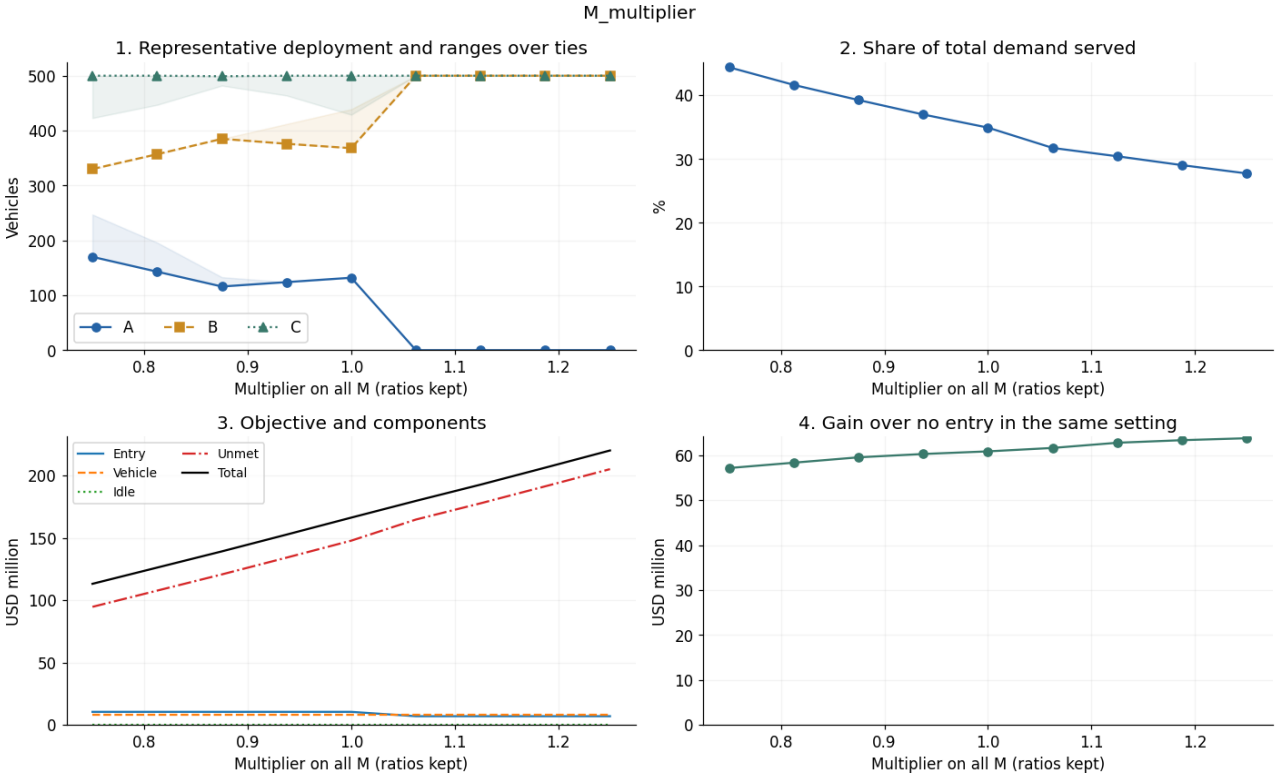


Figure 7: Results from scaling all market sizes with city ratios kept.

Interpretation. Smaller demand keeps three cities and moves vehicles into A (170 at 0.75), because B and C reach their fully served scenarios with fewer vehicles. Larger demand drops A from 1.0625 on: B's cheap tail shrinks as its 3.85-million scenarios grow, so the vehicles A would displace are worth more than A's entry. The switch lies between 1 and 1.0625 (bisection: about 1.042). Note that Z rises with demand even though the improvement rises: the no-entry benchmark Z_0 grows with demand, so Z alone does not rank performance.

3.3.7 City Vehicle Limit $\bar{q}$

Table 19: Selected Results: Common City Vehicle Limit

Value	q^{rep}	Ties	$\bar{T}$	SR (%)	Z	ΔZ
100	$(100, 100, 100)$	1	2,628,000	11.57	213.73	13.39
250	$(250, 250, 250)$	1	6,077,235	26.76	182.82	44.29
500	$(132, 368, 500)$	72	7,932,603	34.93	166.26	60.85
750	$(0, 440, 560)$	1	7,692,016	33.87	165.17	61.95
1,000	$(0, 440, 560)$	1	7,692,016	33.87	165.17	61.95
1,500	$(0, 440, 560)$	1	7,692,016	33.87	165.17	61.95

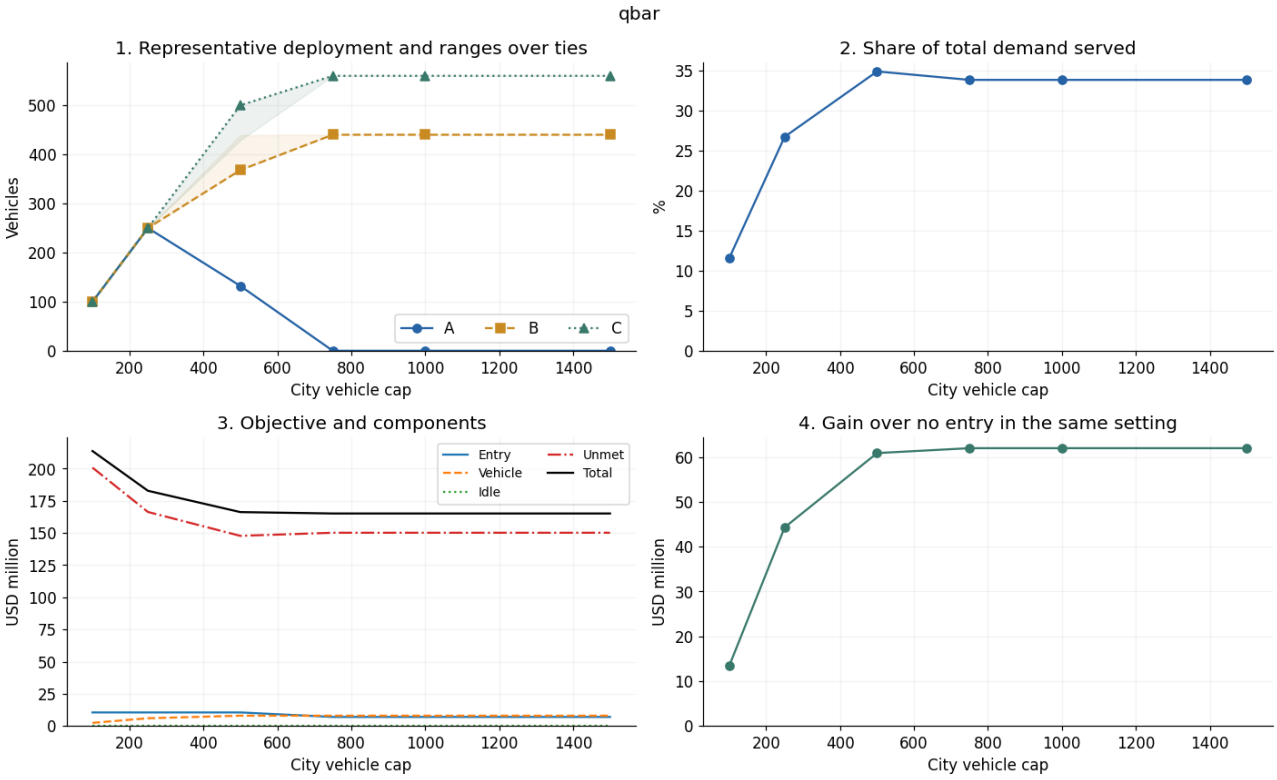


Figure 8: Results from varying the city-specific vehicle limit with $Q = 1{,}000$.

Interpretation. A tight limit forces an even split; at 100 and 250 every city receives its limit. From 750 upward the optimum is $(0, 440, 560)$ and stops changing: C takes 560 vehicles (capacity 4.91 million, just above its 4.82-million scenario), B stops at 440 where its next vehicles would serve only five scenarios, and the extra room in C is worth more than entering A.

3.3.8 Common Unused-Capacity Penalty c^+

Table 20: Selected Results: Common Unused-Capacity Penalty

Value	q^{rep}	Ties	$\bar{T}$	SR (%)	Z	ΔZ
0	$(132, 368, 500)$	72	7,932,603	34.93	166.26	60.85
1	$(132, 368, 500)$	72	7,932,603	34.93	167.09	60.03
3	$(132, 368, 500)$	72	7,932,603	34.93	168.75	58.37
5	$(132, 368, 500)$	72	7,932,603	34.93	170.40	56.72
10	$(132, 368, 500)$	72	7,932,603	34.93	174.54	52.58
15	$(132, 368, 500)$	72	7,932,603	34.93	178.67	48.44
40	$(132, 176, 220)$	1	4,624,827	20.36	195.60	31.52

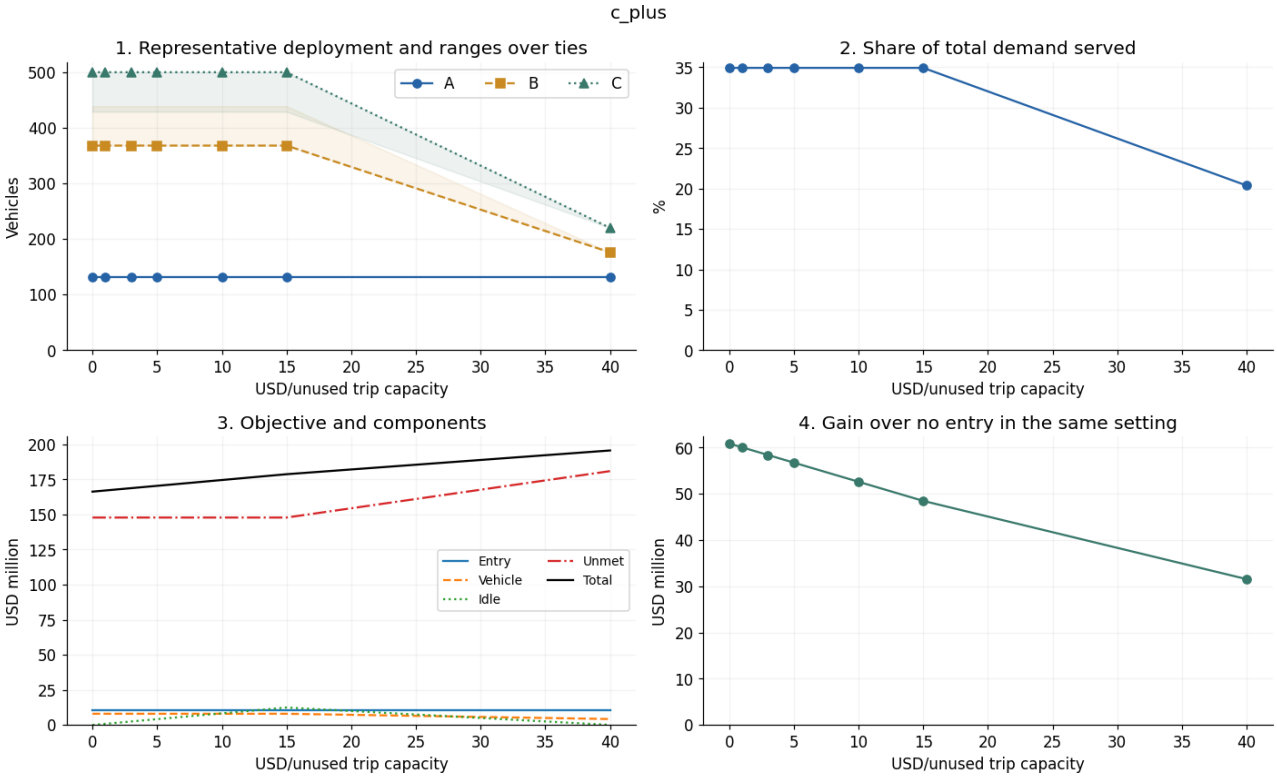


Figure 9: Sensitivity of allocations and performance to the unused-capacity penalty.

Interpretation. From 0 to 15 the allocation is unchanged and $Z = 166{,}263{,}859 + 827{,}397\,c^+$, the baseline objective plus the penalty on the baseline's average unused capacity. The penalty does not reorder cities because it lowers the value of every partially used vehicle in the same way, and the fleet limit keeps all 1,000 vehicles in use. It changes the decision only when a vehicle's value turns negative. A vehicle on the 62,108 step idles $8{,}760 \times 0.2 = 1{,}752$ trips per year on average, so its value is $62{,}108 - 1{,}752\,c^+$, which reaches zero at $c^+ \approx 35.4$. At $c^+ = 40$ the fleet shrinks to $(132, 176, 220)$, exactly the 528 vehicles that are used in every scenario. The penalty can also be absorbed into the other cost parameters: because $O_c^s = \kappa_c q_c - D_c^s + U_c^s$, the objective with (K_c, c_c^-, c_c^+) equals the objective with $(K_c + c_c^+\kappa_c,\ c_c^- + c_c^+,\ 0)$ up to a constant, so c^+ is not an independent decision lever.

3.3.9 Forgone Contribution Margin c^-

Table 21: Selected Results: Common Forgone Contribution Margin

Value	q^{rep}	Ties	$\bar{T}$	SR (%)	Z	ΔZ
2	$(0, 0, 500)$	1	3,889,272	17.12	45.13	0.29
5	$(0, 500, 500)$	1	7,543,306	33.21	90.81	22.74
10	$(132, 368, 500)$	72	7,932,603	34.93	166.26	60.85
15	$(132, 368, 500)$	72	7,932,603	34.93	240.16	100.52
20	$(132, 368, 500)$	72	7,932,603	34.93	314.06	140.18

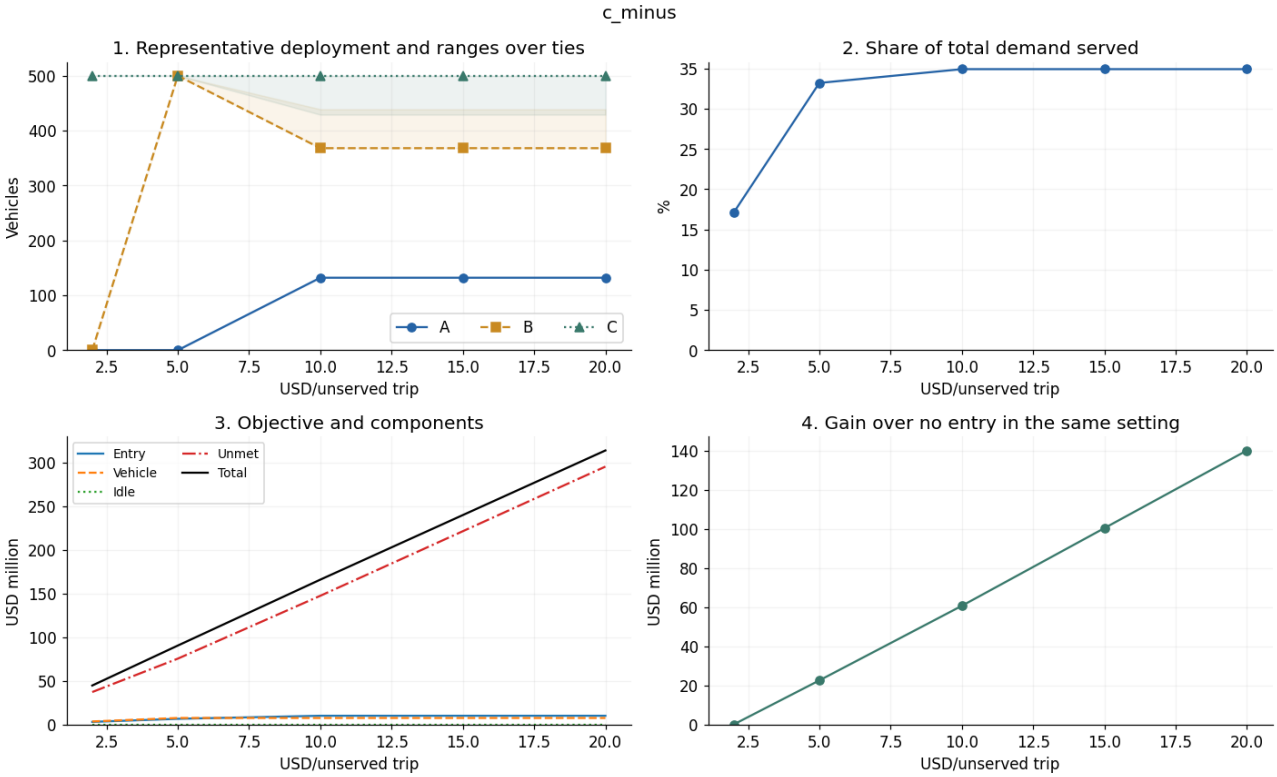


Figure 10: Results from varying the forgone contribution margin.

Interpretation. The margin decides how many entry costs are worth paying. At $c^- = 2$ a vehicle used in every scenario earns $8{,}760 \times 2 - 7{,}972 = 9{,}548$, and only C's 500 vehicles recover an entry cost; at 5 two cities are entered; from 10 upward all three. Above 10 the allocation is fixed and ΔZ grows linearly.

3.3.10 Market Size of A: M_A

Table 22: Selected Results: A's Market-Size Multiplier

Value	q^{rep}	Ties	$\bar{T}$	SR (%)	Z	ΔZ
0.7500	$(0, 500, 500)$	1	7,543,306	35.37	152.79	60.46
0.8125	$(0, 500, 500)$	1	7,543,306	34.81	156.25	60.46
0.8750	$(116, 384, 500)$	56	7,903,275	35.89	159.62	60.56
0.9375	$(124, 376, 500)$	64	7,917,939	35.40	162.94	60.71
1.0000	$(132, 368, 500)$	72	7,932,603	34.93	166.26	60.85
1.0625	$(140, 360, 500)$	80	7,946,770	34.46	169.59	61.00
1.1250	$(149, 351, 500)$	89	7,961,054	34.01	172.91	61.14
1.1875	$(157, 343, 500)$	97	7,975,717	33.58	176.24	61.29
1.2500	$(165, 335, 500)$	105	7,990,381	33.16	179.56	61.43

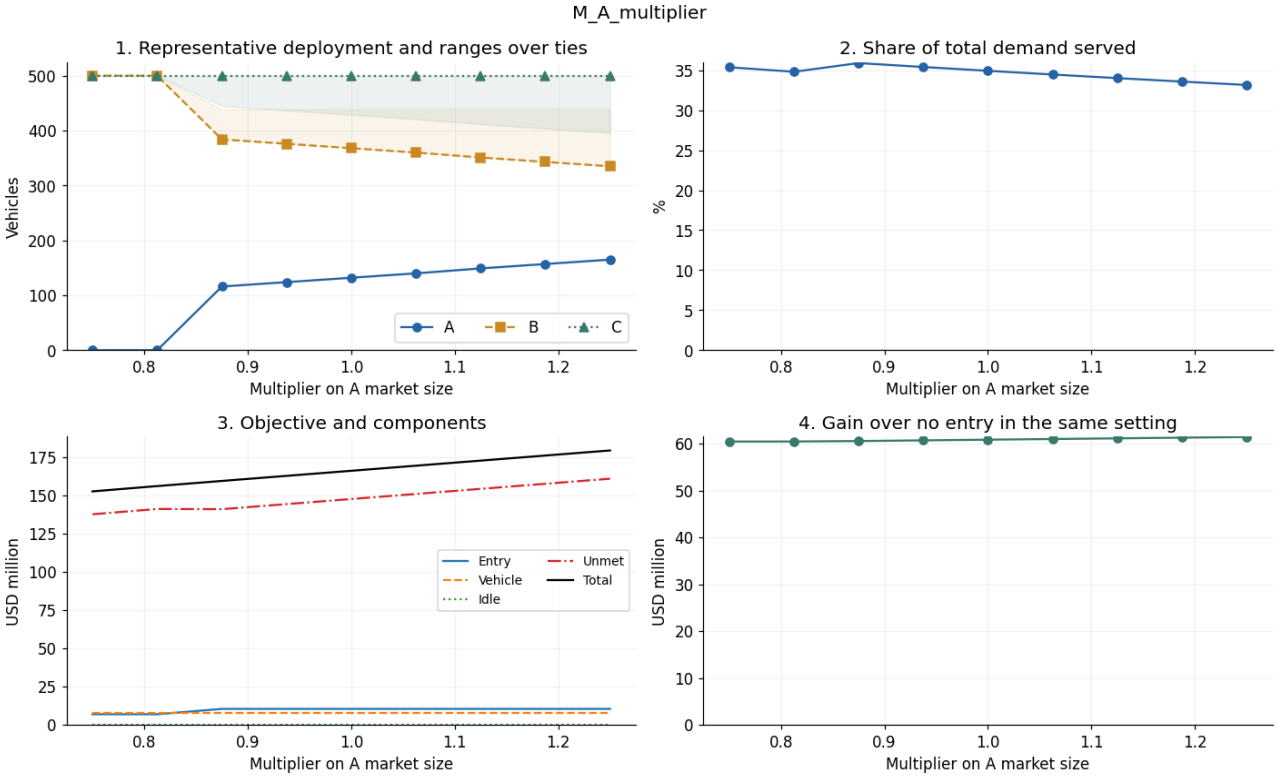


Figure 11: Results from varying only A's market size. A multiplier of 1 corresponds to $M_A = 577{,}908{,}105$.

Interpretation. A's fleet follows its smallest demand scenario: the 132 always-used slots become 116 at 0.875 and 165 at 1.25, and the vehicles come from B. Below 0.875 the always-used slots are too few to repay the entry cost and A is not entered.

3.3.11 Demand Variability α with Fixed Mean Demand

Table 23: Selected Results: Demand Variability

Value	q^{rep}	Ties	$\bar{T}$	SR (%)	Z	ΔZ
0.0	$(0, 500, 500)$	3	8,760,000	38.57	154.49	72.63
0.2	$(0, 500, 500)$	3	8,760,000	38.57	154.49	72.63
0.3	$(0, 500, 500)$	1	8,760,000	38.57	154.49	72.63
0.5	$(0, 500, 500)$	1	8,760,000	38.57	154.49	72.63
0.6	$(0, 500, 500)$	1	8,679,202	38.21	155.30	71.82
0.7	$(0, 500, 500)$	1	8,523,275	37.53	156.86	70.26
0.8	$(232, 315, 453)$	60	8,655,811	38.11	159.03	68.09
0.9	$(182, 318, 500)$	165	8,294,282	36.52	162.65	64.47
1.0	$(132, 368, 500)$	72	7,932,603	34.93	166.26	60.85

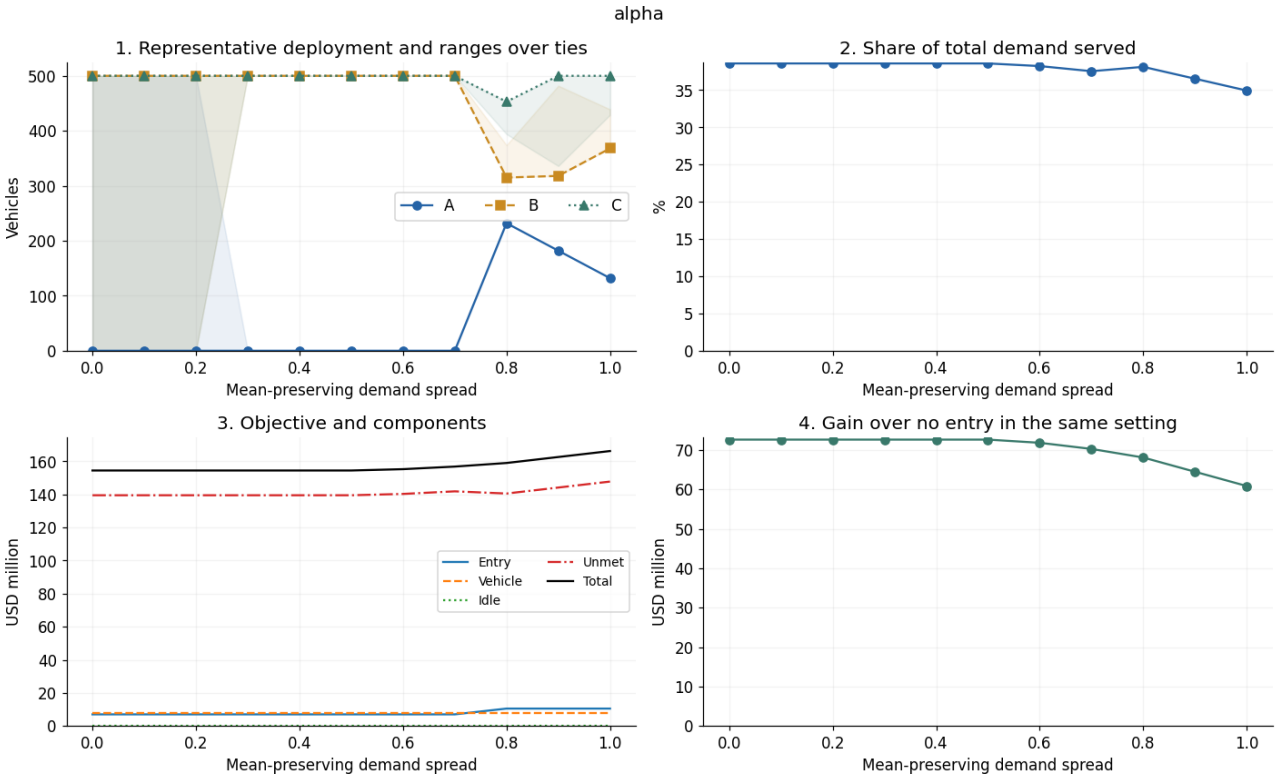


Figure 12: Results from varying demand variability while preserving mean demand.

Interpretation. With little variability every scenario demand exceeds 500 vehicles' capacity, so two cities with 500 vehicles each are optimal and, up to $\alpha = 0.2$, any two cities tie. As variability grows, low-demand scenarios appear in B and C, their late vehicles lose value, and from $\alpha = 0.8$ the fleet spreads to A. The improvement falls from 72.63 to 60.85 million: variability costs money even when the mean is unchanged.

3.4 City-Specific Changes, One at a Time

Each cell changes the listed input in one city only and reports the representative allocation, the number of tied optima in parentheses, and ΔZ in USD million. The baseline is $(132, 368, 500)$ (72) with $\Delta Z = 60.85$.

Table 24: One-City Changes: Representative Allocation (Ties); ΔZ in USD Million

Change	In A	In B	In C
$M_c \times 0.75$	$(0, 500, 500)$ (1); 60.46	$(170, 330, 500)$ (1); 59.75	$(132, 440, 428)$ (16); 59.75
$M_c \times 1.25$	$(165, 335, 500)$ (105); 61.43	$(0, 500, 500)$ (1); 62.81	$(132, 368, 500)$ (72); 61.82
$p_c^s \times 0.75$	$(0, 500, 500)$ (1); 60.46	$(170, 330, 500)$ (1); 59.75	$(132, 440, 428)$ (16); 59.75
$p_c^s \times 1.25$	$(165, 335, 500)$ (105); 61.43	$(0, 500, 500)$ (1); 62.81	$(132, 368, 500)$ (72); 61.82
$\kappa_c \times 0.75$	$(0, 500, 500)$ (1); 60.46	$(266, 234, 500)$ (1); 55.57	$(267, 440, 293)$ (1); 54.53
$\kappa_c \times 1.25$	$(264, 236, 500)$ (204); 63.74	$(148, 352, 500)$ (1); 66.88	$(132, 368, 500)$ (1); 68.95
$F_c \times 0.75$	$(132, 368, 500)$ (72); 61.73	$(132, 368, 500)$ (72); 61.73	$(132, 368, 500)$ (72); 61.73
$F_c \times 1.25$	$(0, 500, 500)$ (1); 60.46	$(132, 368, 500)$ (72); 59.98	$(132, 368, 500)$ (72); 59.98
$K_c \times 0.75$	$(132, 368, 500)$ (72); 61.12	$(132, 439, 429)$ (1); 61.73	$(132, 368, 500)$ (1); 61.85
$K_c \times 1.25$	$(132, 368, 500)$ (72); 60.59	$(132, 368, 500)$ (1); 60.12	$(132, 439, 429)$ (1); 60.00
$c_c^- \times 0.75$	$(0, 500, 500)$ (1); 60.46	$(324, 176, 500)$ (1); 55.32	$(330, 440, 230)$ (1); 54.12
$c_c^- \times 1.25$	$(330, 176, 494)$ (264); 65.04	$(0, 500, 500)$ (1); 69.60	$(132, 368, 500)$ (1); 70.58
$c_c^+ = 1$	$(132, 368, 500)$ (72); 60.85	$(132, 368, 500)$ (1); 60.52	$(132, 439, 429)$ (1); 60.49
$c_c^+ = 3$	$(132, 368, 500)$ (72); 60.85	$(132, 368, 500)$ (1); 59.84	$(132, 440, 428)$ (1); 59.76
$c_c^+ = 5$	$(132, 368, 500)$ (72); 60.85	$(132, 368, 500)$ (193); 59.17	$(132, 440, 428)$ (198); 59.03
$\bar{q}_c = 100$	$(0, 500, 500)$ (1); 60.46	$(400, 100, 500)$ (1); 55.95	$(400, 500, 100)$ (61); 53.59
$\bar{q}_c = 1{,}000$	$(132, 368, 500)$ (72); 60.85	$(132, 368, 500)$ (72); 60.85	$(0, 440, 560)$ (1); 61.95

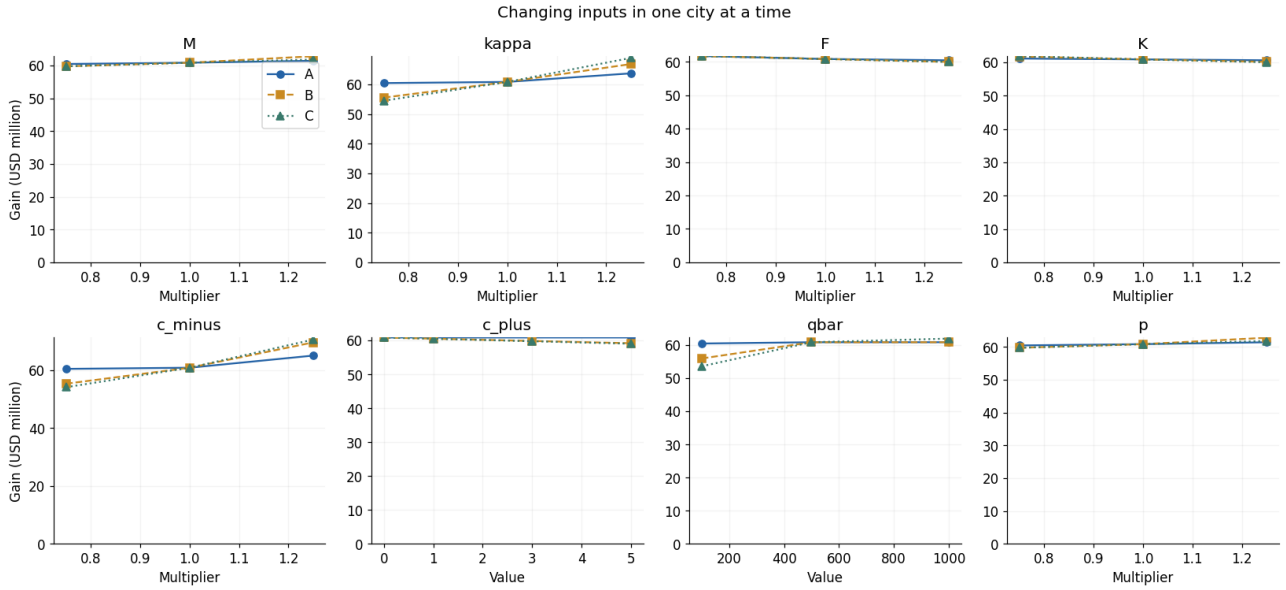


Figure 13: Improvement over no entry when one input is changed in one city.

Interpretation. A cheaper or more productive city attracts vehicles: lower F or K, or higher κ or c^-, in one city pulls the fleet toward it, and the opposite change pushes the fleet away. Unlike the reference-level run, one city's M or p now matters too: a 25% larger market in B removes A, while a 25% smaller market in C moves vehicles from C to B. Small city-specific costs act as tie breakers: an unused-capacity penalty of 1 in B alone makes $(132, 368, 500)$ the unique optimum, and the same penalty in C alone selects $(132, 439, 429)$, because idle capacity now costs money in one city only. A common change that leaves the allocation unchanged therefore says nothing about robustness to differences between cities.

3.5 Summary

Table 25: Where the Optimal Allocation Changes (Baseline $(132, 368, 500)$)

Input	Range examined	Allocation change
Q	0–2,000	C only up to 500; B added at 600; A added at 1,000; limits bind at 1,500
F (common)	±25%	A dropped above 3,892,967
K (common)	±25%	None; ΔZ changes linearly
κ (common)	±25%	A dropped at $\leq$ 8,212.5 (switch near 8,498)
M or p (common)	×0.75–1.25	A dropped at $\geq$ 1.0625 (switch near 1.042); more vehicles in A when demand is smaller
$\bar{q}$	100–1,500	Even split at 100, 250; $(0, 440, 560)$ from 750
c^+ (common)	0–40	None up to 15; fleet shrinks to 528 at 40 (switch near 35.4)
c^- (common)	2–20	One city at 2, two at 5, three from 10
M_A	×0.75–1.25	A dropped at $\leq$ 0.8125; q_A from 116 to 165 otherwise
α	0–1	Two cities up to 0.7; three cities from 0.8
One-city changes	±25%; c^+ 1–5; $\bar{q}$ 100, 1,000	Vehicles move toward the cheaper or more productive city; single-city c^+ breaks the B–C tie

Three points carry over to the modeling discussion. First, city-specific demand differences affect the allocation only when demand and capacity are of the same order; when demand dominates, cost parameters alone decide. Second, with ten equally likely scenarios and $c^+ = 0$ the value of a vehicle is a step function, which produces large sets of tied optima; more scenarios or a positive c^+ reduce them. Third, the switching points reported here belong to this synthetic instance and are not estimates for real cities.